User's Manual

Qdt User's Manual for FC7xxx Series

Rev: 1.2.0

Target Devices:

FC7300F8MDQ, FC7300F8MDT

FC7300F4MDD, FC7300F4MDS

FC7240F2MDS

Flagchip Semiconductors

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Chapter 1. Qdt Introduction

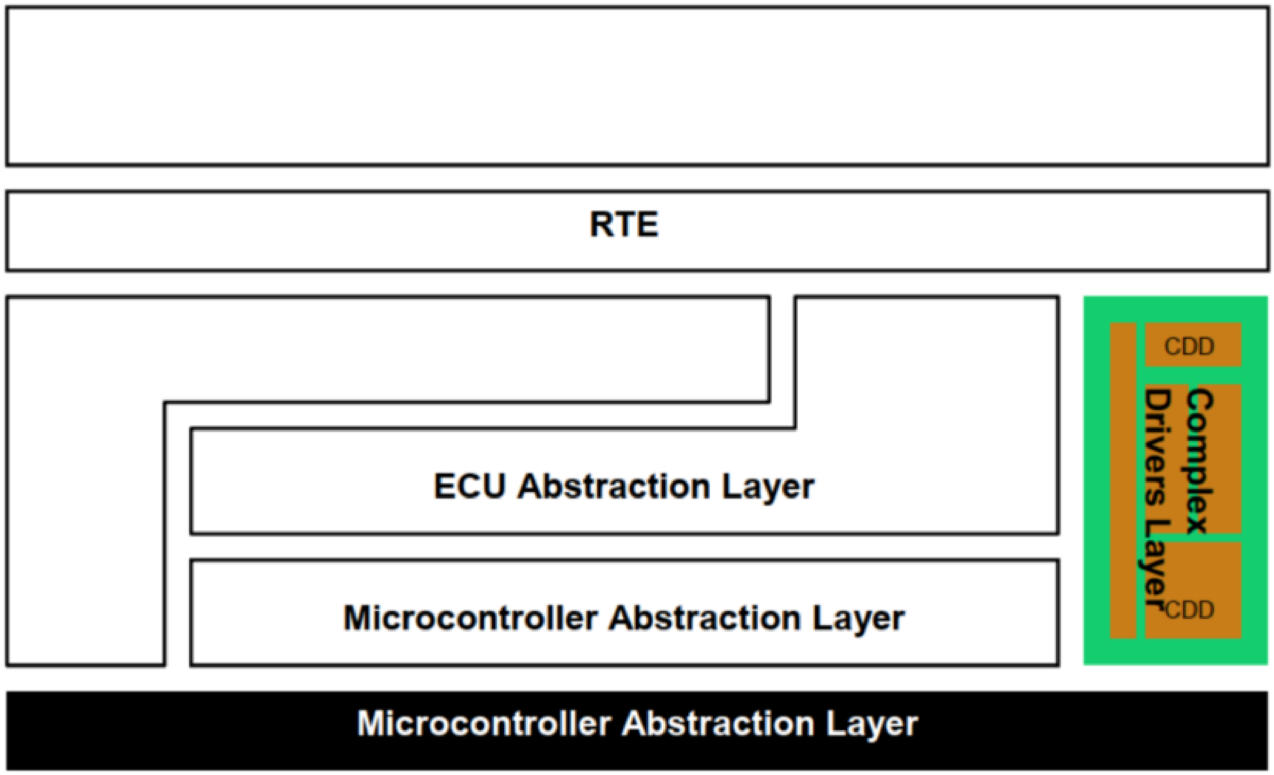
1.1. Requirements

The design of this module follows the specifications of the complex driver specified in AUTOSAR Classic Platform Release R20-11. For detailed requirements, refer to the *AUTOSAR_EXP_CDDDesignAndIntegrationGuideline.*

1.2. Introduction to CDD

A Complex Driver is a software entity not standardized by AUTOSAR that can access or be accessed via AUTOSAR Interfaces and/or Basic Software Modules APIs.According to the autosar Layered Software Architecture, a CDD is a specific module located in the Complex Drivers Layer of the Basic Software which interacts with standard BSW modules or Rte.

- A CDD may need to interface to modules of the layered software architecture
- A module of the layered software architecture may need to interface to a CDD
- A CDD may need to interface SW-Cs via RTE



1.3. Hardware Summary

The QDT driver is responsible to initialize and control the Quadrature Decoder Timer module. The QDT driver provides user interfaces to configure the QDT parameters.

The QDT module has 4 instances in FC7300F8MDQ, and each instance also has 4 channels.The following table lists the Quadrature Decoder Timer (QDT) instances of each chip in the product series and summarizes features that vary across the products and instances.

Chips	Instances	Channels	Quadrate
FC7300F8MDQ	QDT0	4	Yes
	QDT1	4	Yes
	QDT2	4	Yes
	QDT3	4	Yes

QDT module input modes:

- Input Capture (IC) Mode
- Input Capture Measurement (ICM) Mode
  - Input Capture Duty-Cycle Measurement (ICDM) Mode
  - Input Capture Period Measurement (ICPM) Mode
  - Input Capture Edge Number Measurement (ICENM) Mode
  - Input Capture Expect Edge Number Measurement (ICEXPENM) Mode
- Quadrature Decoder (QUAD) Mode

QDT module features:

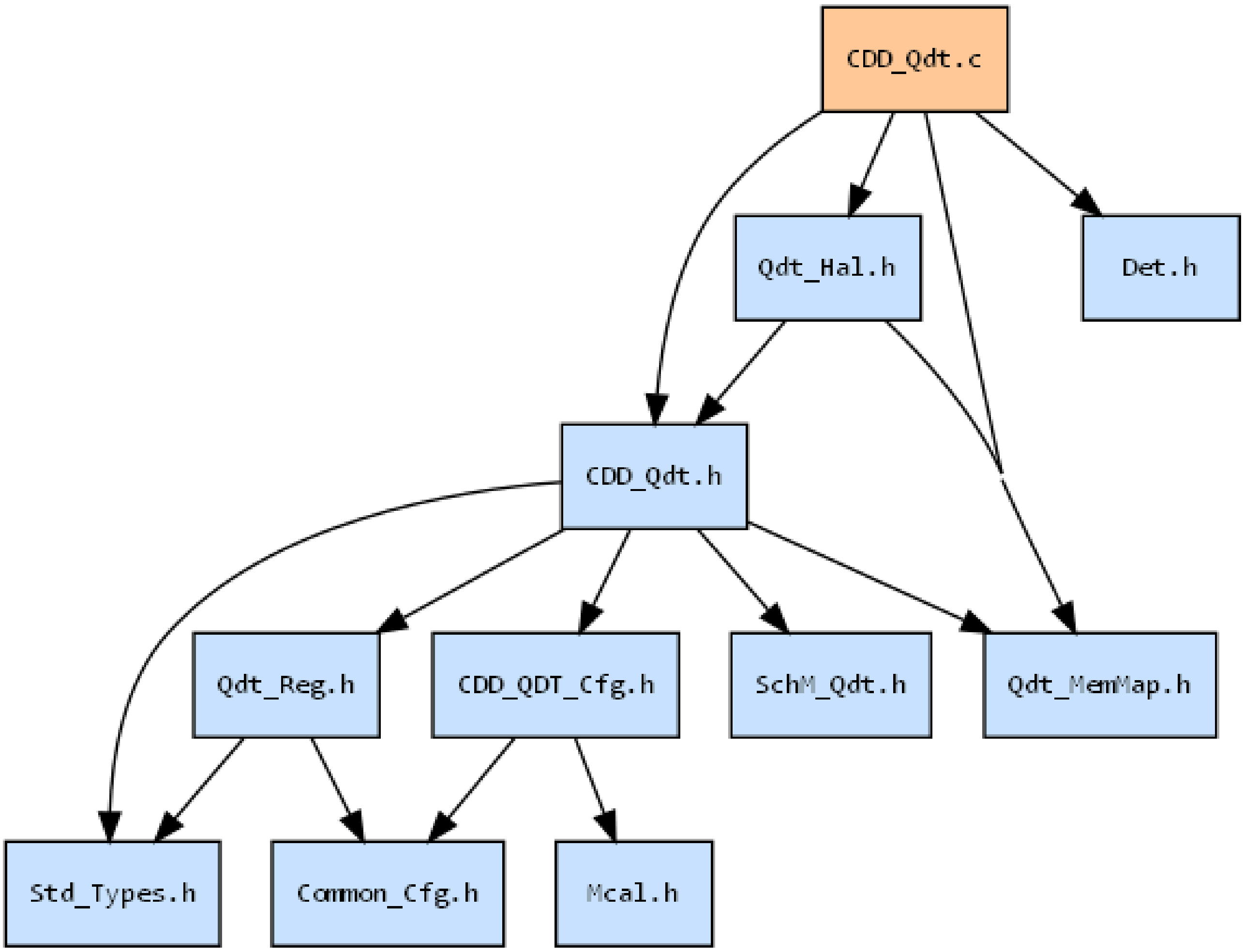
- QDT source clock is selectable
- Prescaler divided by 1, 2, 4, 8, 16, 32, 64, or 128 for input mode and watch dog
- Generation trigger of match point
- Polarity of Phase A/B input is configurable
- Generation of an interrupt per channel
- Synchronized loading of write buffered QDT registers
- Register reload capacity
- Write protection for critical registers
- Direct access to input pin states
- Each channel can be configured for input capture, quadrature decoder mode
- Measurement of duty-cycle, period, edge number, expect edge number of input signal
- Quadrature decoder with input glitch filters, relative position counting, and interrupt on position count or capture of position count on an external event
- The quadrature decoder can decode shaft position, revolution count, and speed
- 32-bit position counter
- Generation of an interrupt when position counter overflows
- 16-bit position difference counter
- 16-bit revolution counter
- A watchdog timer can detect a non-rotating shaft condition

Chapter 2. Software Design

2.1. Rejected Requirements

N/A

2.2. File Description



2.3. Define and Macro reference

2.3.1. Macros in CDD_Qdt.h

#define QDT_VENDOR_ID 174

#define QDT_MODULE_ID 255

#define QDT_AR_RELEASE_MAJOR_VERSION 4

#define QDT_AR_RELEASE_MINOR_VERSION 6

#define QDT_AR_RELEASE_REVISION_VERSION 0

#define QDT_SW_MAJOR_VERSION 1

#define QDT_SW_MINOR_VERSION 2

#define QDT_SW_PATCH_VERSION 0

#define QDT_E_ALREADY_UNINITIALIZED_U8 (uint8)0x0AU

#define QDT_E_INVALID_CHANNEL_U8 (uint8)0x0BU

#define QDT_E_UNINIT_U8 (uint8)0x0CU

#define QDT_E_ALREADY_INITIALIZED_U8 (uint8)0x0DU

#define QDT_E_PARAM_U8 (uint8)0x0EU

#define QDT_E_INIT_FAILED_U8 (uint8)0x0FU

#define QDT_E_PARTITION_MAPPING (uint8)0x10U

#define QDT_INIT_ID_U8 (uint8)0x1U

#define QDT_DEINIT_ID_U8 (uint8)0x2U

#define QDT_GETVERSIONINFO_ID_U8 (uint8)0x3U

#define QDT_RESTART_MEASUREMENT_ID_U8 (uint8)0x4U

#define QDT_GET_CHANNEL_FLAG_ID_U8 (uint8)0x5U

#define QDT_CLEAR_CHANNEL_FLAG_ID_U8 (uint8)0x6U

#define QDT_GET_EDGE_NUMBER_ID_U8 (uint8)0x7U

#define QDT_GET_CV_ID_U8 (uint8)0x8U

#define QDT_GET_REVCT_ID_U8 (uint8)0x9U

#define QDT_GET_REVCNT_HOLD_ID_U8 (uint8)0xAU

#define QDT_GET_POSCNT_ID_U8 (uint8)0xB

#define QDT_GET_POSCNT_HOLD_ID_U8 (uint8)0xC

#define QDT_GET_POSDCNT_ID_U8 (uint8)0xD

#define QDT_GET_POSDCNT_HOLD_ID_U8 (uint8)0xE

#define QDT_GET_LECNT_ID_U8 (uint8)0xF

#define QDT_GET_LECNT_HOLD_ID_U8 (uint8)0x10U

#define QDT_GET_POSDTMRCNT_ID_U8 (uint8)0x11U

#define QDT_GET_POSDTMRCNT_HOLD_ID_U8 (uint8)0x12U

#define QDT_GET_SPEED_ID_U8 (uint8)0x13U

2.4. Enum reference

2.4.1. Enums in CDD_Qdt.handle

2.4.1.1. QDT_StatusType

Enumeration	QDT_StatusType	
Description	QDT status type.	
Values	**Value**	**Description**
	QDT_UNINIT	The QDT is not initialized.
	QDT_INITIALIZED	The QDT has been initialized.

2.4.1.2. QDT_ReturnType

Enumeration	QDT_ReturnType	
Description	QDT operation return values.	
Values	**Value**	**Description**
	QDT_RETURN_OK	The QDT operation succeeded.
	QDT_RETURN_E_NOT_OK	The QDT operation failed.
	QDT_RETURN_E_ALREADY_INIT	The QDT has been initialized.
	QDT_RETURN_E_UNINIT	The QDT is not initialized.
	QDT_RETURN_E_PARAM	The QDT parameter is incorrect or out of range.
	QDT_RETURN_E_SEQUENCE	Sequence error.
	QDT_RETURN_E_MODE	Mode error.

2.4.1.3. QDT_ClockSourceType

Enumeration	QDT_ClockSourceType	
Description	QDT clock source type.	
Values	**Value**	**Description**
	QDT_CLOCK_NONE	No clock selected.
	QDT_CLOCK_INTERNAL_BUSCLK	QDT input clock from bus clock.
	QDT_CLOCK_INTERNAL_PCCCLK	QDT input clock from pcc clock. PCC_SEL is not 0.
	QDT_CLOCK_EXTERNAL_TCLK0	External pin input clock from QDT_TCLK0 pin. Only valid when PCC_SEL is 0.
	QDT_CLOCK_EXTERNAL_TCLK1	External pin input clock from QDT_TCLK1 pin. Only valid when PCC_SEL is 0.
	QDT_CLOCK_EXTERNAL_TCLK2	External pin input clock from QDT_TCLK2 pin. Only valid when PCC_SEL is 0.

2.4.1.4. QDT_CounterPrescalerType

Enumeration	QDT_CounterPrescaleType	
Description	QDT counter prescaler.	
Values	**Value**	**Description**
	QDT_COUNTER_PRESCALE_DIV_1	QDT counter clock frequency divide by 1.
	QDT_COUNTER_PRESCALE_DIV_2	QDT counter clock frequency divide by 2.
	QDT_COUNTER_PRESCALE_DIV_4	QDT counter clock frequency divide by 4.
	QDT_COUNTER_PRESCALE_DIV_8	QDT counter clock frequency divide by 8.
	QDT_COUNTER_PRESCALE_DIV_16	QDT counter clock frequency divide by 16.
	QDT_COUNTER_PRESCALE_DIV_32	QDT counter clock frequency divide by 32.
	QDT_COUNTER_PRESCALE_DIV_64	QDT counter clock frequency divide by 64.
	QDT_COUNTER_PRESCALE_DIV_128	QDT counter clock frequency divide by 128.

2.4.1.5. QDT_ChannelIndexType

Enumeration	QDT_ChannelIndexType	
Description	QDT channel index.	
Values	**Value**	**Description**
	QDT_CHANNEL_0_PHA	QDT channel index 0, PHA pad.
	QDT_CHANNEL_1_PHB	QDT channel index 1, PHB pad.
	QDT_CHANNEL_2_PHZ	QDT channel index 2, PHZ pad.
	QDT_CHANNEL_3_HOME	QDT channel index 3, HOME pad.

2.4.1.6. QDT_ChannelModeType

Enumeration	QDT_ChannelModeType	
Description	QDT channel operation mode.	
Values	**Value**	**Description**
	QDT_CHANNEL_NOT_USED	QDT channel is not used.
	QDT_CHANNEL_IC_MODE	QDT channel is in IC mode.
	QDT_CHANNEL_ICDM_MODE	QDT channel is in ICDM mode.
	QDT_CHANNEL_ICPM_MODE	QDT channel is in ICPM mode.
	QDT_CHANNEL_ICENM_MODE	QDT channel is in ICENM mode.
	QDT_CHANNEL_ICEXPENM_MODE	QDT channel is in ICEXPENM mode.
	QDT_CHANNEL_QUAD_MODE	QDT channel is in QUAD mode.

2.4.1.7. QDT_CaptureEdgeType

Enumeration	QDT_CaptureEdgeType	
Description	QDT capture edge type.	
Values	**Value**	**Description**
	QDT_CAPTURE_RISING_EDGE	Capture the rising edge.
	QDT_CAPTURE_FALLING_EDGE	Capture the falling edge.
	QDT_CAPTURE_BOTH_EDGE	Capture both edges.

2.4.1.8. QDT_ChannelMatchType

Enumeration	QDT_ChannelMatchType	
Description	QDT channel match type.	
Values	**Value**	**Description**
	QDT_CHANNEL_MATCH_POSCNT	Match point is POSCNT.
	QDT_CHANNEL_MATCH_REVCNT	Match point is REVCNT.
	QDT_CHANNEL_MATCH_DISABLE	Disable the match trigger.

2.5. Structures and typedefs reference

2.5.1. QDT_ConfigType

Data structure containing the set of configuration parameters required for initializing the Qdt instance(s) and channel(s).

```
typedef struct
{
    uint8                 u8InstanceCount;
    QDT_InstanceConfigType  **pQdtInstanceCfg;
    boolean*               Qdt_CoresMappingPtr;
    uint32*                Qdt_CtrlCoresMappingPtr;
} QDT_ConfigType;
```

Data Fields

Type	Member	Description
uint8	u8InstanceCount	The counter of the used Qdt instance.
QDT_InstanceConfigType **	pAdcPtr	The Pointer array of the QDT_InstanceConfigType.
boolean *	Qdt_CoresMappingPtr	The Pointer array of the core mapping, if true, means the current core need to manage at least one qdt instance.
uint32 **	Qdt_CtrlCoresMappingPtr	The Pointer array of the controller mapping, the value indicate which core should manage the qdt instance.

2.5.2. QDT_InstanceConfigType

QDT instance configuration.

```
typedef struct
{
    uint8                   u8InstanceLogicID;
    uint8                   u8InstanceHwIndex;
    uint32                  u32BusClockFreq;
  uint32                  u32LecntLarge;
  uint32                  u32EncoderLine;
    boolean                 bEnPHAPHBMode;
    boolean                 bEnQuadMode;
    boolean                 bEnDebugMode;
    boolean                 bEnMatchPulse;
    QDT_ClockSourceType        eClkSrcSel;
    QDT_CounterPrescaleType     eCounterPrescale;
  uint8                   u8FilterPrescale;
    boolean                 bEnTOFInt;
    QDT_CallbackType          pTOFCallback;
    QDT_WdogConfigType     const *pWdgConfig;
    QDT_SyncModeConfigType const *pSyncModeConfig;
    uint8                   u8ChannelCount;
    QDT_ChannelConfigType     **pQdtChannelCfg;
}QDT_InstanceConfigType;
```

Data Fields

Type	Member	Description
uint8	u8InstanceLogicID	QDT instance logic ID.
uint8	u8InstanceHwIndex	QDT instance hardware index.
uint32	u32BusClockFreq	The frequency of Bus clock.
uint32	u32LecntLarge	The max lecnt value.
uint32	u32EncoderLine	Line number of encoder.
boolean	bEnPHAPHBMode	true: Phase A and phase B encoding mode. false: Count and direction encoding mode.
boolean	bEnQuadMode	QUAD Mode Enable. false is disable, true is enable.
boolean	bEnDebugMode	Debug Mode Enable. false is disable, true is enable.
boolean	bEnMatchPulse	true means Match Trigger pulses when a match occurs between the position counters (POS) and the corresponding channel value (CV) false means Match Trigger pulses when the POSCNT, REVCNT, or POSDCNT are read.
QDT_ClockSourceType	eClkSrcSel	Select the QDT clock source.
QDT_CounterPrescaleType	eCounterPrescale	Setting the prescale of QDT counter clock frequency.

Type	Member	Description
uint8	u8FilterPrescale	Setting the prescale of QDT input filter.
boolean	bEnTOFInt	Timer Overflow Interrupt Enable. true is disable, false is enable.
QDT_CallbackType	pTOFCallback	Timer Overflow Interrupt callback function.
QDT_WdogConfigType *	pWdgConfig	QDT watchdog configuration.
QDT_SyncModeConfigType *	pSyncModeConfig	QDT sync mode configuration.
uint8	u8ChannelCount	The counter of channel used.
QDT_ChannelConfigType **	pQdtChannelCfg	The Pointer array of the Qdt_ChannelConfigType.

2.5.3. QDT_SyncModeConfigType

QDT sync mode configuration.

```
typedef struct
{
   boolean   bEnCVSyncTriggerMode;  /*!<  */
   boolean   bEnSoftTriggerReset;   /*!<  */
   boolean   bEnHardTriggerInput;   /*!<  */
   boolean   bEnHardTriggerReset;   /*!<  */
   boolean   bEnHardTriggerUpdate;  /*!<  */
}QDT_SyncModeConfigType;
```

Data Fields

Type	Member	Description
boolean	bEnSoftTriggerReset	True: Sync CV register with trigger mode.
boolean	u8InstanceHwIndex	True: Allow SW event to reset POSCNT, REVCNT and POSDCNT. FALSE: SW event will only reset the POSCNT.
boolean	bEnHardTriggerInput	Enables hardware trigger to the synchronization and reset.
boolean	bEnHardTriggerReset	True: Allow HW event to reset POSCNT, REVCNT and POSDCNT. FALSE: HW event will not reset any Counter.
boolean	bEnHardTriggerUpdate	Allow the hardware trigger to cause an update of the POSCNTH, REVCNTH and POSDCNTH registers.

2.5.4. QDT_WdogConfigType

QDT wdog configuration.

```
typedef struct
{
   boolean    bEnWDOG;
   boolean    bEnWDOGFInt;
   uint16     u16Timeout;
   QDT_CallbackType pWDOGFCallback;
}QDT_WdogConfigType;
```

Data Fields

Type	Member	Description
boolean	bEnWDOG	Enable/Disable the watchdog feature.
boolean	bEnWDOGFInt	Enable/Disable the watchdog timeout interrupt.
uint16	u16Timeout	The value of the watchdog timeout.
QDT_CallbackType	pWDOGFCallback	Watchdog Timer Overflow Interrupt callback function.

2.5.5. QDT_ChannelConfigType

QDT channel configuration.

```
typedef struct
{
    uint8              u8LogicChannelID;
    uint8              u8HwChannelIndex;
    uint8              u8InputFilter;
    QDT_ChannelModeType    eChannelMode;

    boolean              bEnChannelInt;
    QDT_CallbackType       pChannelCallback;
    union
    {
        QDT_IC_ConfigType          const *pICConfig;
        QDT_ICDM_ConfigType        const *pICDMConfig;
        QDT_ICPM_ConfigType        const *pICPMConfig;
        QDT_ICENM_ConfigType       const *pICENMConfig;
        QDT_ICEXPENM_ConfigType    const *pICEXPENMConfig;
        QDT_QUAD_ConfigType        const *pQUADConfig;
    };
}QDT_ChannelConfigType;
```

Data Fields

Type	Member	Description
uint8	u8LogicChannelID	Logic Qdt Channel ID.
uint8	u8HwChannelIndex	Hardware Qdt Channel ID.
uint8	u8InputFilter	Channel input filter. 0 means disable the filter.
QDT_ChannelModeType	eChannelMode	Select Channel operation mode.
boolean	bEnChannelInt	Enable the channel interrupt.
QDT_CallbackType	pChannelCallback	Channel interrupt Notification.
QDT_IC_ConfigType *	pICConfig	Pointer of IC mode configuration.
QDT_ICDM_ConfigType *	pICDMConfig	Pointer of ICDM mode configuration.
QDT_ICPM_ConfigType *	pICPMConfig	Pointer of ICPM mode configuration.
QDT_ICENM_ConfigType *	pICENMConfig	Pointer of ICENM mode configuration.
QDT_ICEXPENM_ConfigType *	pICEXPENMConfig	Pointer of ICEXPENM mode configuration.
QDT_QUAD_ConfigType *	pQUADConfig	Pointer of QUAD mode configuration.

2.5.6. QDT_QUAD_ConfigType

QDT QUAD mode configuration.

```
typedef struct
{
  boolean   bNormalPolarity;
  uint32    u32CV;
  QDT_ChannelMatchType  eMatchType;
}QDT_QUAD_ConfigType;
```

Data Fields

Type	Member	Description
boolean	u8InstanceCount	true: Phase A and phase B encoding mode. false: Count and direction encoding mode
uint32	u32CV	CV is used to match event. Compare with the source selected by MatchType.
QDT_ChannelMatchType	eMatchType	Select the match type for CV.

2.5.7. QDT_ICEXPENM_ConfigType

QDT ICEXPENM mode configuration.

```
typedef struct
{
    boolean             bContinueMeasure;
    QDT_CaptureEdgeType eEdge;
    uint8               u8ExpectedNum;
}QDT_ICEXPENM_ConfigType;
```

Data Fields

Type	Member	Description
boolean	bContinueMeasure	indicates whether continue to measure.
QDT_CaptureEdgeType	eEdge	Select which edge to capture.
uint8	u8ExpectedNum	The number of expected edge.

2.5.8. QDT_ICENM_ConfigType

QDT ICENM mode configuration.

```
typedef struct
{
    boolean             bContinueMeasure;
    QDT_CaptureEdgeType  eEdge;
    uint32              u32StartPoint;
    uint32              u32EndPoint;
}QDT_ICENM_ConfigType;
```

Data Fields

Type	Member	Description
boolean	u8InstanceCount	indicates whether continue to measure.
QDT_CaptureEdgeType	eEdge	Select which edge to capture.
uint32	u32StartPoint	The start of the measure window.
uint32	u32EndPoint	The end of the measure window.

2.5.9. QDT_ICPM_ConfigType

QDT ICPM mode configuration.

```
typedef struct
{
    boolean              bContinueMeasure;
    QDT_CaptureEdgeType      eEdge;
    boolean              bStartWIthActive;
}QDT_ICPM_ConfigType;
```

Data Fields

Type	Member	Description
boolean	bContinueMeasure	indicates whether continue to measure.
QDT_CaptureEdgeType	eEdge	Select which edge to capture.please don't select both edge.
boolean	bStartWIthActive	TRUE: the channel starts measuring after the first edge is detected.FALSE: the measurement starts immediately after activating the channel.

2.5.10. QDT_ICDM_ConfigType

QDT ICDM mode configuration.

```
typedef struct
{
    boolean      bContinueMeasure;
    boolean      bHighActive;
    boolean      bStartWIthActive;
}QDT_ICDM_ConfigType;
```

Data Fields

Type	Member	Description
boolean	bContinueMeasure	indicates whether continue to measure.
boolean	bHighActive	Indicate high /low valid of input signal.
boolean	bStartWIthActive	TRUE: the channel starts measuring after the first edge is detected.FALSE: the measurement starts immediately after activating the channel.

2.5.11. QDT_IC_ConfigType

QDT IC mode configuration.

```
typedef struct
{
   QDT_CaptureEdgeType    eEdge;
   boolean                bFromRevCNT;
   boolean                bResetPOSCNT;
}QDT_IC_ConfigType;
```

Data Fields

Type	Member	Description
QDT_CaptureEdgeType	eEdge	Select which edge to capture.
boolean	bFromRevCNT	FALSE: capture POSCNT into the CV register, TRUE: capture REVCNT into the CV register.
boolean	bResetPOSCNT	FALSE:QDT counter is not reset when the selected channel (n) input event is detected.

2.6. Function reference

2.6.1. QDT_Init

Function that initializes the QDT module.

```
void QDT_Init(const QDT_ConfigType *pConfigPtr)
```

Parameters

Type	Name	Direction	Description
QDT_ConfigType *	pConfigPtr	input	pointer to QDT configuration structure.

Returns

void

> **Notification**
>
> The function Autosar Service ID[hex]: 0x01.

> **Notification**
>
> Synchronous.

> **Notification**
>
> Non Reentrant function.

2.6.2. QDT_DeInit

This function de-initializes the QDT driver.

```
void QDT_DeInit(void)
```

Parameters

void

Returns

void

> 🔔 **Notification**
>
> The function Autosar Service ID[hex]: 0x02.

> 🔔 **Notification**
>
> Synchronous.

> 🔔 **Notification**
>
> Non Reentrant function.

2.6.3. QDT_GetVersionInfo

This service returns the version information of this module.

```
void QDT_GetVersionInfo(Std_VersionInfoType *pVersioninfo)
```

Parameters

Type	Name	Direction	Description
Std_VersionInfoType *	pVersioninfo	output	Pointer to location to store version info.

Returns

void

> 🔔 **Notification**
>
> The function Autosar Service ID[hex]: 0x03.

> 🔔 **Notification**
>
> Synchronous.

Notification

Non Reentrant function.

2.6.4. QDT_ReStartMeasurement

This service Re-start a measurement in Single Measurement Mode.

QDT_ReturnType QDT_ReStartMeasurement(uint8 u8ControllerID, uint8 u8ChannelID)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint8	u8ChannelID	input	The qdt channel index

Returns

QDT_ReturnType

Notification

The function Autosar Service ID[hex]: 0x04.

Notification

Synchronous.

Notification

Non Reentrant function.

2.6.5. QDT_GetChannelFlag

This service get the channel flag, used for polling method.

QDT_ReturnType QDT_GetChannelFlag(uint8 u8ControllerID, uint8 u8ChannelID, boolean *pSet)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint8	u8ChannelID	input	The qdt channel index.
boolean *	pSet	output	The pointer of getting if the flag is set.

Returns

QDT_ReturnType

Notification

The function Autosar Service ID[hex]: 0x05.

Notification

Synchronous.

Notification

Non Reentrant function.

2.6.6. QDT_ClearChannelFlag

This service clear the channel flag, used for polling method.

QDT_ReturnType QDT_ClearChannelFlag(uint8 u8ControllerID, uint8 u8ChannelID)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint8	u8ChannelID	input	The qdt channel index.

Returns

QDT_ReturnType

Notification

The function Autosar Service ID[hex]: 0x06.

Notification

Synchronous.

Notification

Non Reentrant function.

2.6.7. QDT_GetCV

This service get the CV counter value.

QDT_ReturnType QDT_GetCV(uint8 u8ControllerID, uint8 u8ChannelID, uint32 *pCV)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint8	u8ChannelID	input	The qdt channel index.
uint32 *	pCV	output	The pointer of getting the CV counter value.

Returns

QDT_ReturnType

Notification

The function Autosar Service ID[hex]: 0x07.

Notification

Synchronous.

Notification

Non Reentrant function.

2.6.8. QDT_GetREVCNT

This service get the REV counter value.

QDT_ReturnType QDT_GetREVCNT(uint8 u8ControllerID, uint32 *pREVCNT)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint32 *	pREVCNT	output	The pointer of getting the REV counter value.

Returns

QDT_ReturnType

Notification

The function Autosar Service ID[hex]: 0x08.

Notification

Synchronous.

Notification

Non Reentrant function.

2.6.9. QDT_GetREVCNT_HOLD

This service get the REV HOLD counter value.

QDT_ReturnType QDT_GetREVCNT_HOLD(uint8 u8ControllerID, uint32 *pREVCNTH)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint32 *	pREVCNTH	output	The pointer of getting the REV HOLD counter value.

Returns

QDT_ReturnType

Notification

The function Autosar Service ID[hex]: 0x09.

Notification

Synchronous.

Notification

Non Reentrant function.

2.6.10. QDT_GetPOSCNT

This service get the POS counter value.

QDT_ReturnType QDT_GetPOSCNT(uint8 u8ControllerID, uint32 *pPOSCNT)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint32 *	pPOSCNT	output	The pointer of getting the POS counter value.

Returns

QDT_ReturnType

> **Notification**
>
> The function Autosar Service ID[hex]: 0x0A.

> **Notification**
>
> Synchronous.

> **Notification**
>
> Non Reentrant function.

2.6.11. QDT_ResetPOSCNT

This service reset the POS counter value.

QDT_ReturnType QDT_ResetPOSCNT(uint8 u8ControllerID)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.

Returns

QDT_ReturnType

> **Notification**
>
> The function Autosar Service ID[hex]: 0x0A.

> **Notification**
>
> Synchronous.

> **Notification**
>
> Non Reentrant function.

2.6.12. QDT_GetPOSCNT_HOLD

This service get the POS HOLD counter value.

QDT_ReturnType QDT_GetPOSCNT_HOLD(uint8 u8ControllerID, uint32 *pPOSCNTH)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint32 *	pPOSCNTH	output	The pointer of getting the POS HOLD counter value.

Returns

QDT_ReturnType

> **Notification**
>
> The function Autosar Service ID[hex]: 0x0B.

> **Notification**
>
> Synchronous.

> **Notification**
>
> Non Reentrant function.

2.6.13. QDT_GetPOSDCNT

This service get the POS DIFF counter value.

QDT_ReturnType QDT_GetPOSDCNT(uint8 u8ControllerID, uint32 *pPOSDCNT)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint32 *	pPOSDCNT	output	The pointer of getting the POS DIFF counter value.

Returns

QDT_ReturnType

Notification

The function Autosar Service ID[hex]: 0x0C.

Notification

Synchronous.

Notification

Non Reentrant function.

2.6.14. QDT_GetPOSDCNT_HOLD

This service get the POS DIFF HOLD counter value.

QDT_ReturnType QDT_GetPOSDCNT_HOLD(uint8 u8ControllerID, uint32 *pPOSDCNTH)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint32 *	pPOSDCNTH	output	The pointer of getting the POS DIFF HOLD counter value.

Returns

QDT_ReturnType

Notification

The function Autosar Service ID[hex]: 0x0D.

Notification

Synchronous.

Notification

Non Reentrant function.

2.6.15. QDT_GetLECNT

This service get the LE counter value.

QDT_ReturnType QDT_GetLECNT(uint8 u8ControllerID, uint32 *pLECNT)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint32 *	pLECNT	output	The pointer of getting the LE counter value.

Returns

QDT_ReturnType

Notification

The function Autosar Service ID[hex]: 0x0E.

Notification

Synchronous.

Notification

Non Reentrant function.

2.6.16. QDT_GetLECNT_HOLD

This service get the LE HOLD counter value.

QDT_ReturnType QDT_GetLECNT_HOLD(uint8 u8ControllerID, uint32 *pLECNTH)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint32 *	pLECNTH	output	The pointer of getting the LE HOLD counter value.

Returns

QDT_ReturnType

Notification

The function Autosar Service ID[hex]: 0x0F.

Notification

Synchronous.

Notification

Non Reentrant function.

2.6.17. QDT_GetPOSDTMRCNT

This service get the POSDTMR counter value.

QDT_ReturnType QDT_GetPOSDTMRCNT(uint8 u8ControllerID, uint32 *pPOSDTMRCNT)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint32 *	pPOSDTMRCNT	output	The pointer of getting the POSDTMR counter value.

Returns

QDT_ReturnType

Notification

The function Autosar Service ID[hex]: 0x10.

Notification

Synchronous.

Notification

Non Reentrant function.

2.6.18. QDT_GetPOSDTMRCNT_HOLD

This service get the POSDTMR HOLD counter value.

QDT_ReturnType QDT_GetPOSDTMRCNT_HOLD(uint8 u8ControllerID, uint32 *pPOSDTMRCNTH)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
uint32 *	pPOSDTMRCNTH	output	The pointer of getting the POSDTMR HOLD counter value.

Returns

QDT_ReturnType

Notification

The function Autosar Service ID[hex]: 0x11.

Notification

Synchronous.

Notification

Non Reentrant function.

2.6.19. QDT_GetSpeed

This service get the speed from the sensor.

QDT_ReturnType QDT_GetSpeed(uint8 u8ControllerID, float *pSpeed)

Parameters

Type	Name	Direction	Description
uint8	u8ControllerID	input	The qdt controller index.
float *	pSpeed	output	The pointer of getting the speed value.

Returns

QDT_ReturnType

Notification

The function Autosar Service ID[hex]: 0x12.

Notification

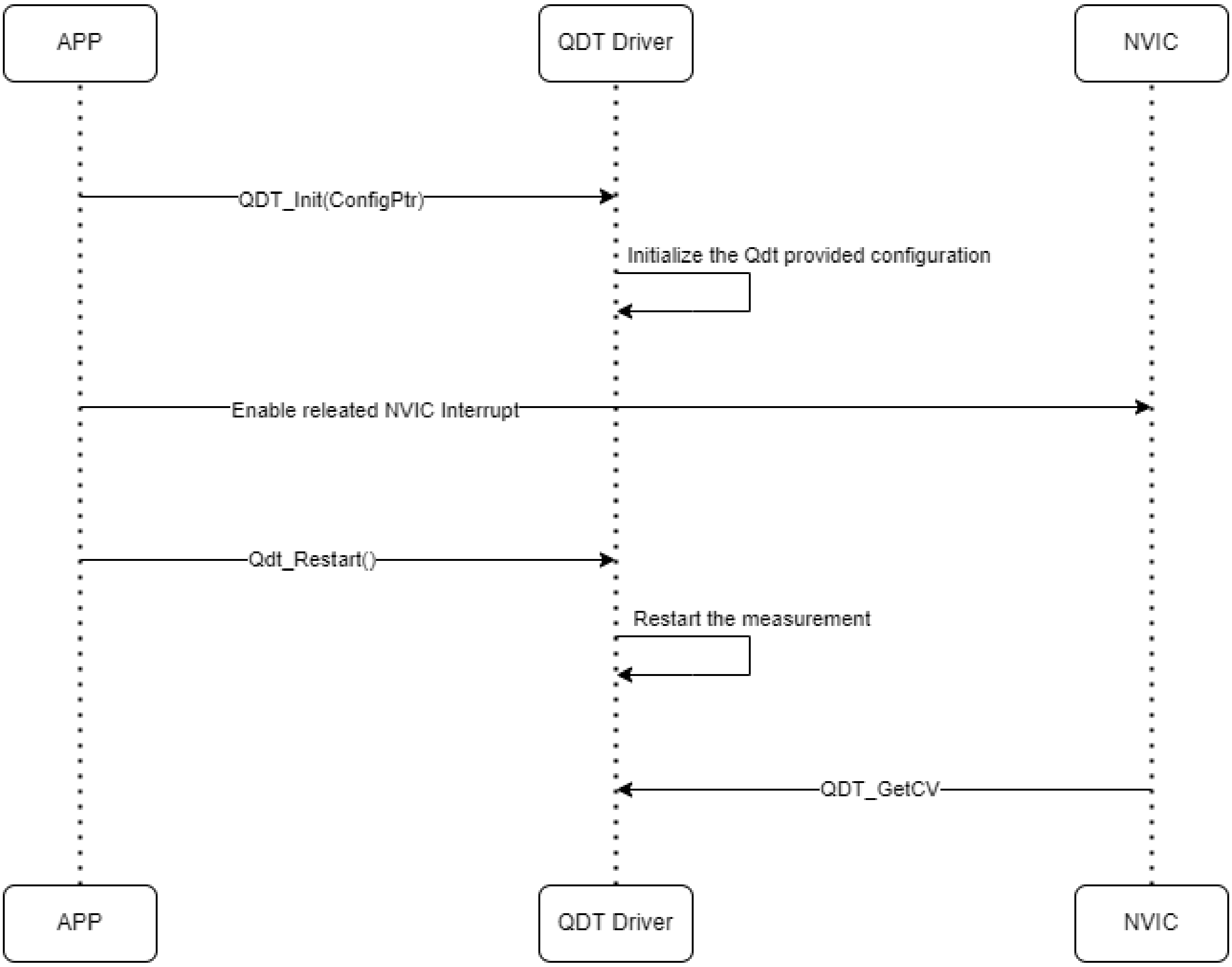
Synchronous.

Notification

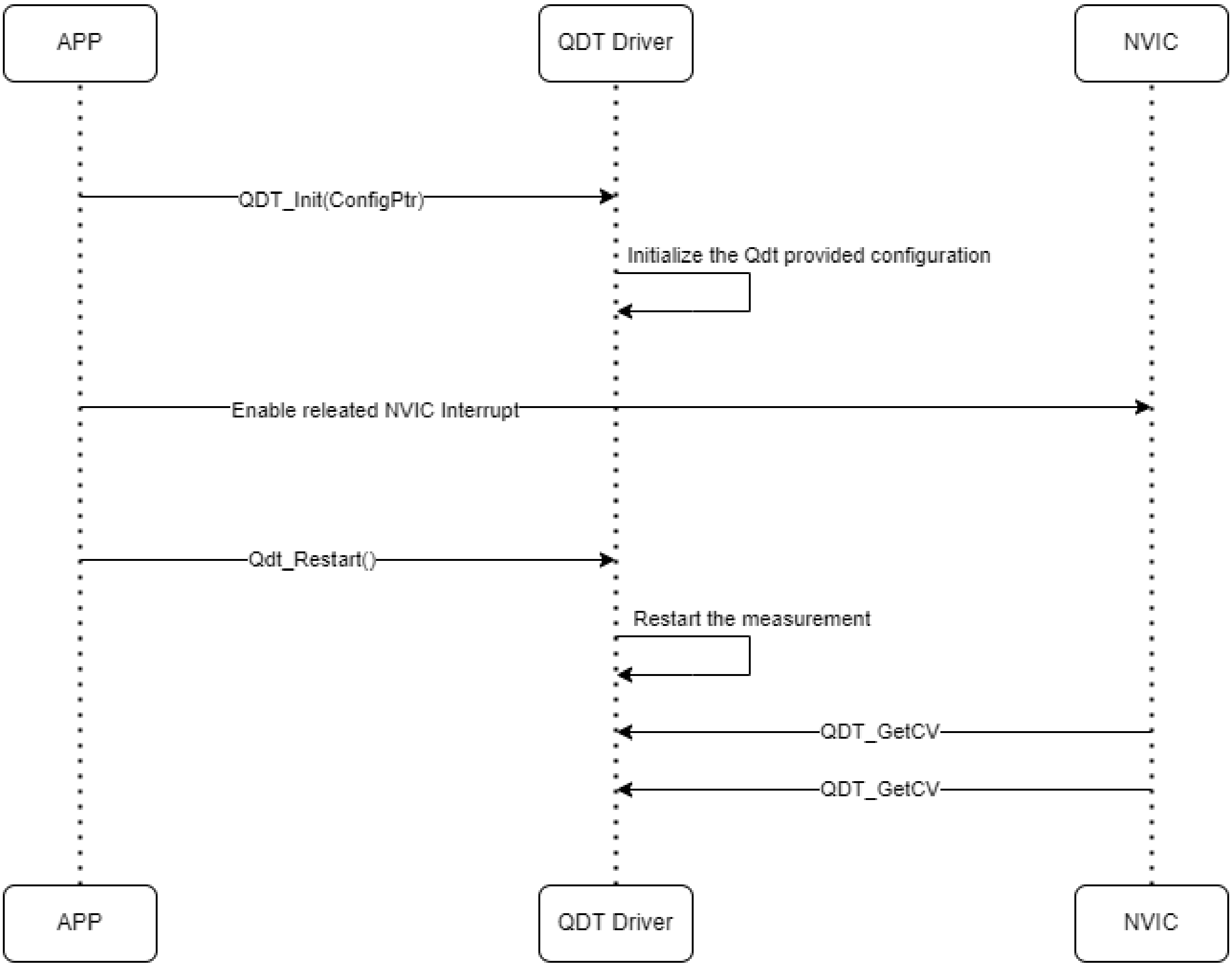
Non Reentrant function.

2.7. API Sequence Diagram

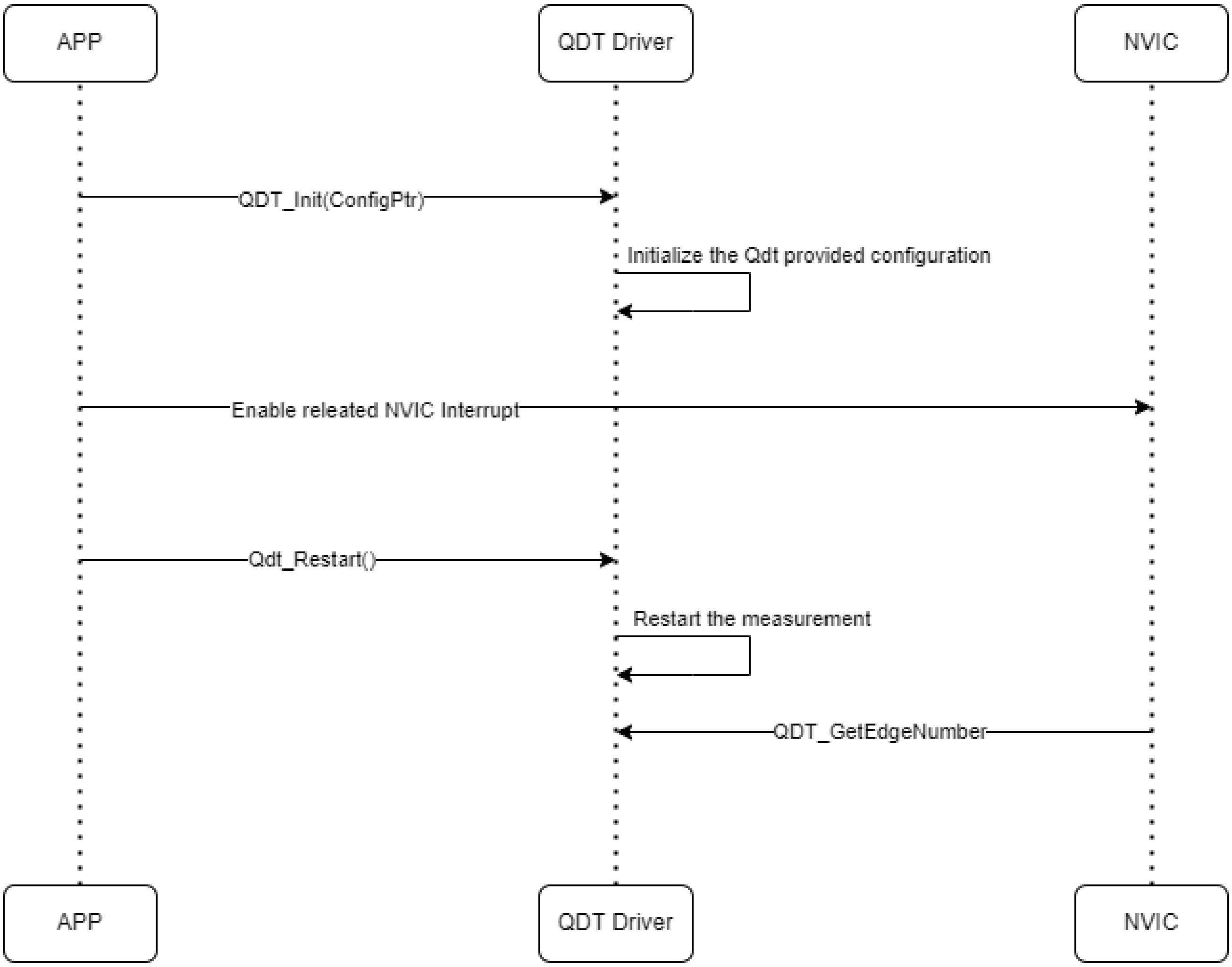
2.7.1. IC Mode



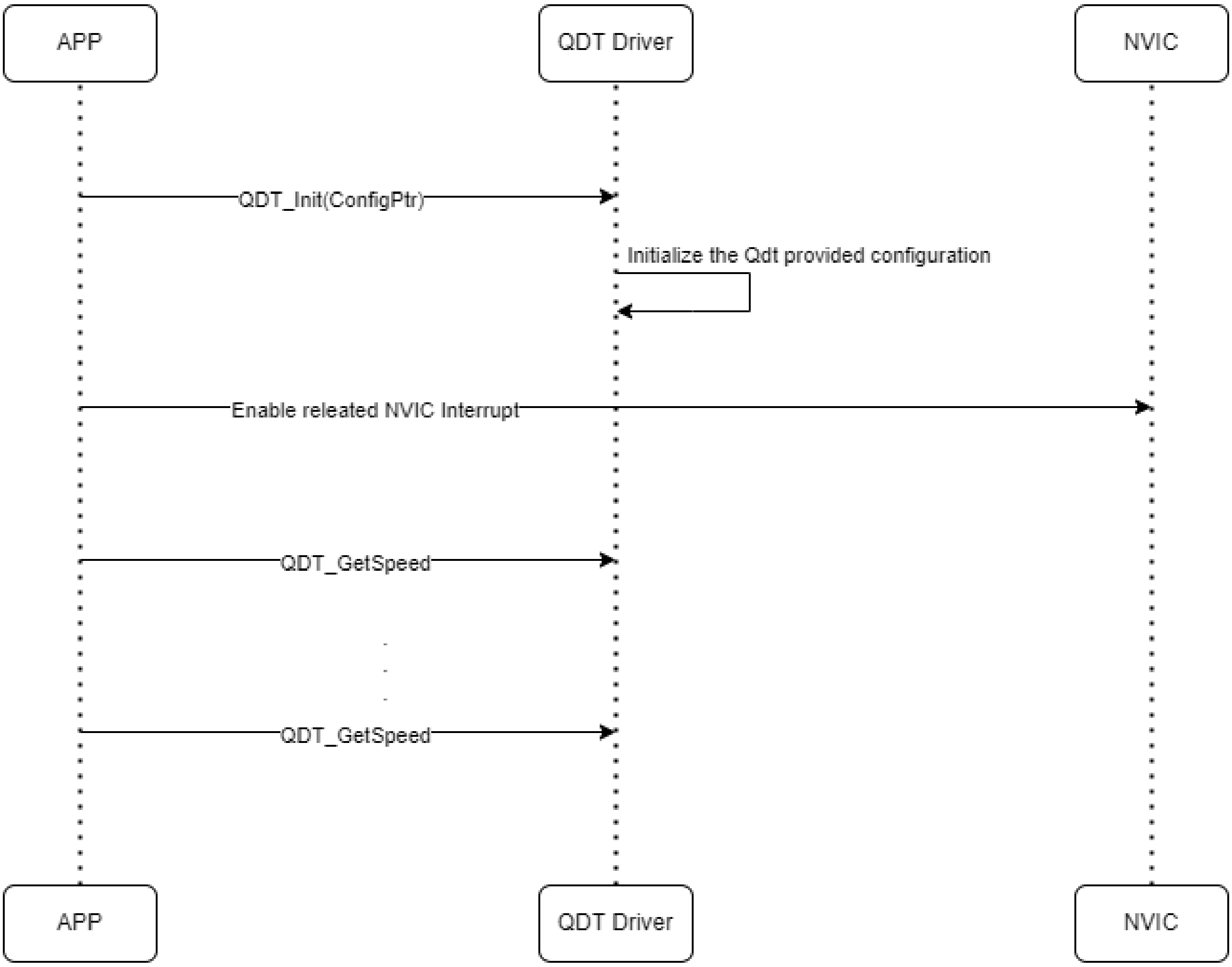
2.7.2. ICDM Mode & ICEXPENM Mode & ICPM Mode



2.7.3. ICENM Mode

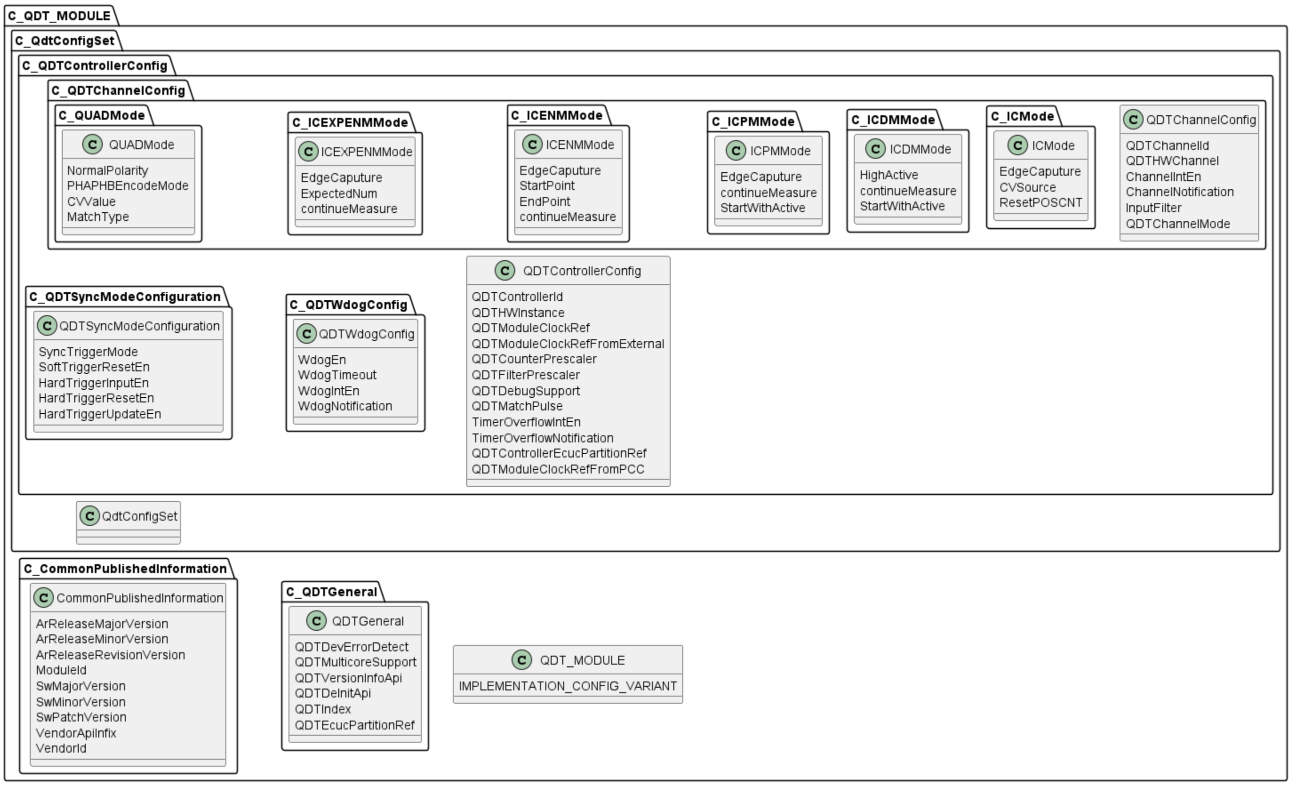


2.7.4. QUAD Mode



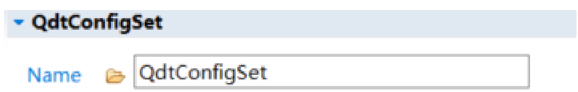
Chapter 3. Tresos Configuration Items

3.1. Container Inclusion Relation

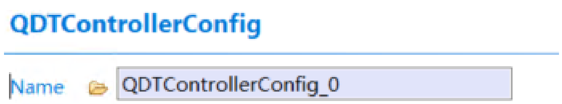


3.2. Containers and Variables

3.2.1. QdtConfigSet

Container	QdtConfigSet	
Description	This is the multiple configuration set container for QDT Driver.	
Screenshot		
Properties	**Property**	Value
	Type	IDENTIFIABLE

3.2.1.1. QDTControllerConfig

Container	QDTControllerConfig	
Description	Contains the channels of each QDT controller	
Screenshot		
Properties	**Property**	Value
	Type	IDENTIFIABLE

3.2.1.1.1. QDTControllerId

Container	QDTControllerId	
Description	This parameter provides the controller ID which is unique in a given QDT Driver. The value for this parameter starts with 0 and continue without any gaps.	
Screenshot	QDT Controller ID	0
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	true
	Default	None
	Origin	Flagchip

3.2.1.1.2. QDTHWInstance

Container	QDTHWInstance	
Description	Specifies which one of the on-chip QDT interfaces is associated with this controller ID.	
Screenshot	QDT hardware Instance	Qdt0
Properties	**Property**	Value
	Type	ENUMERATION
	Symbolic Name	false
	Default	None
	Origin	Flagchip

3.2.1.1.3. QDTModuleClockRef

Container	QDTModuleClockRef	
Description	Select the QDT function clock source QDT_CLOCK_INTERNAL_BUSCLK: QDT function clock from bus clock. QDT_CLOCK_INTERNAL_PCCCLK: QDT function clock from pcc clock.	
Screenshot	QDT fcuntion clock source	QDT_CLOCK_INTERNAL_PCCCLK
Properties	**Property**	Value
	Type	ENUMERATION
	Symbolic Name	false
	Default	QDT_CLOCK_INTERNAL_BUSCLK
	Origin	Flagchip

3.2.1.1.4. QDTModuleClockRefFromExternal

Container	QDTModuleClockRefFromExternal	
Description	Select the external clock as function clock source.	
Screenshot	fcuntion clock source from externl pin	QDT_CLOCK_EXTERNAL_TCLK1

Properties	**Property**	Value
	Type	ENUMERATION
	Symbolic Name	none
	Default	QDT_CLOCK_EXTERNAL_TCLK0
	Origin	Flagchip

3.2.1.1.5. QDTCounterPrescaler

Container	QDTCounterPrescaler	
Description	Select the prescale of QDT counter clock frequency.	
Screenshot	QDT counter prescaler QDT_COUNTER_PRESCALE_DIV_8	
Properties	**Property**	Value
	Type	ENUMERATION
	Symbolic Name	none
	Default	QDT_COUNTER_PRESCALE_DIV_1
	Origin	Flagchip

3.2.1.1.6. QDTFilterPrescaler

Container	QDTFilterPrescaler	
Description	1 to 16 - Setting the prescale of QDT filter.	
Screenshot	Time Stamp Prescaller (1 -> 16) 1	
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	1
	Origin	Flagchip

3.2.1.1.7. QDTDebugSupport

Container	QDTDebugSupport	
Description	Debug Mode Enable.	
Screenshot	QDT Debug Mode QDT match pulse	
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.8. QDTMatchPulse

Container	QDTMatchPulse

Description	true means Match Trigger pulses when a match occurs between the position counters (POS) and the corresponding channel value(CV). false means Match Trigger pulses when the POSCNT, REVCNT, or POSDCNT are read.	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.9. TimerOverflowIntEn

Container	TimerOverflowIntEn	
Description	When the QDT counter (also is position counter) passes the value in the MOD register, generate the interrupt	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.10. TimerOverflowNotification

Container	TimerOverflowNotification	
Description	This parameter defines the existence and the name of a callout function that is called after timer overflow occured. No call should occur if this is not enabled.	
Screenshot		
Properties	**Property**	Value
	Type	FUNCTION-NAME
	Symbolic Name	false
	Default	NULL_PTR
	Origin	Flagchip

3.2.1.1.11. QDTControllerEcucPartitionRef

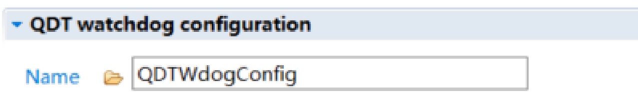
Container	QDTControllerEcucPartitionRef	
Description	EN: Maps a QDT controller to zero or multiple ECUC partitions to limit the access to this controller. The ECUC partitions referenced are a subset of the ECUC partitions where the QDT driver is mapped to.	
Screenshot		
Properties	**Property**	Value

	Type	FUNCTION-REFERENCE
	Symbolic Name	none
	Default	none
	Origin	Flagchip

3.2.1.1.12. QDTModuleClockRefFromPCC

Container	QDTModuleClockRefFromPCC	
Description	Reference to the PCC clock configuration, which is set in the MCU driver configuration. MCU plugin need to be added and then give the reference to it.	
Screenshot		
Properties	**Property**	Value
	Type	FUNCTION-REFERENCE
	Symbolic Name	none
	Default	none
	Origin	Flagchip

3.2.1.1.13. QDTWdogConfig

Container	QDTWdogConfig	
Description	EN: This container contains the configuration (parameters) of the QDT watchdog.	
Screenshot		
Properties	**Property**	Value
	Type	FUNCTION-IDENTIFIABLE

3.2.1.1.13.1. WdogEn

Container	WdogEn	
Description	EN: Enable/Disable the watchdog feature.	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN-REFERENCE
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.13.2. WdogTimeout

Container	WdogTimeout
Description	EN: WDOG Timeout Value
Screenshot	

Properties	**Property**	Value
	Type	INTEGER-REFERENCE
	Symbolic Name	false
	Default	65535
	Origin	Flagchip

3.2.1.1.13.3. WdogIntEn

Container	WdogIntEn	
Description	EN: WDOG Timerout Interrupt Enable	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN-REFERENCE
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.13.4. WdogNotification

Container	WdogNotification	
Description	Set here the name of the handler for QDT watchdog timeout notification function.	
Screenshot		
Properties	**Property**	Value
	Type	FUNCTION-NAME
	Symbolic Name	false
	Default	NULL_PTR
	Origin	Flagchip

3.2.1.1.14. QDTSyncModeConfiguration

Container	QDTSyncModeConfiguration	
Description	EN: This container contains the configuration (parameters) of the QDT sync mode.	
Screenshot		
Properties	**Property**	Value
	Type	IDENTIFIABLE-IDENTIFIABLE

3.2.1.1.14.1. SyncTriggerMode

Container	SyncTriggerMode
Description	EN: True: Sync CV register with trigger mode.

Screenshot	QDT sync with trigger mode	
Properties	**Property**	Value
	Type	BOOLEAN-REFERENCE
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.14.2. SoftTriggerResetEn

Container	SoftTriggerResetEn	
Description	EN: True: Allow SW event to reset POSCNT, REVCNT and POSDCNT. FALSE: SW event will only reset the POSCNT.	
Screenshot	software tigger to reset QDT counter	
Properties	**Property**	Value
	Type	BOOLEAN-REFERENCE
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.14.3. HardTriggerInputEn

Container	HardTriggerInputEn	
Description	EN: Enables hardware trigger 0 to the synchronization and reset.	
Screenshot	enable the hardware trigger input	
Properties	**Property**	Value
	Type	BOOLEAN-REFERENCE
	Symbolic Name	false
	Default	false
	Origin	Flagchip

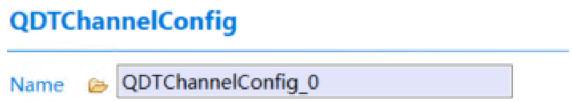
3.2.1.1.14.4. HardTriggerResetEn

Container	HardTriggerResetEn	
Description	EN: True: Allow HW event to reset POSCNT, REVCNT and POSDCNT. FALSE: HW event will not reset any Counter.	
Screenshot	hardware tigger to reset QDT counter	
Properties	**Property**	Value
	Type	BOOLEAN-REFERENCE
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.14.5. HardTriggerUpdateEn

Container	HardTriggerUpdateEn	
Description	EN: Allow the hardware trigger to cause an update of the POSCNTH, REVCNTH and POSDCNTH registers.	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN-REFERENCE
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.15. QDTChannelConfig

Container	QDTChannelConfig	
Description	This container contains the configuration (parameters) of the QDT channels.	
Screenshot		
Properties	**Property**	Value
	Type	IDENTIFIABLE

3.2.1.1.15.1. QDTChannelId

Container	QDTChannelId	
Description	This parameter provides the Channel ID which is unique in a given QDT channel. The value for this parameter starts with 0 and continue without any gaps.	
Screenshot		
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	true
	Default	none
	Origin	Flagchip

3.2.1.1.15.2. QDTHWChannel

Container	QDTHWChannel	
Description	Specifies which one of the on-chip QDT interfaces is associated with this controller ID.	
Screenshot		
Properties	**Property**	Value

	Type	ENUMERATION
	Symbolic Name	false
	Default	none
	Origin	Flagchip

3.2.1.1.15.3. ChannelIntEn

Container	ChannelIntEn	
Description	EN: QDT channel Interrupt Enable	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.15.4. ChannelNotification

Container	ChannelNotification	
Description	Set here the name of the handler for QDT channel notification function.	
Screenshot		
Properties	**Property**	Value
	Type	FUNCTION-NAME
	Symbolic Name	false
	Default	NULL_PTR
	Origin	Flagchip

3.2.1.1.15.5. InputFilter

Container	InputFilter	
Description	Selects the filter value for current channel input. The filter is disabled when the value is zero. The filter value is the value*4	
Screenshot		
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	0
	Origin	Flagchip

3.2.1.1.15.6. QDTChannelMode

Container	QDTChannelMode
Description	EN: Select the channel operation mode.

Screenshot	QDT channel mode QDT_CHANNEL_IC_MODE	
Properties	**Property**	Value
	Type	ENUMERATION
	Symbolic Name	none
	Default	QDT_CHANNEL_NOT_USED
	Origin	Flagchip

3.2.1.1.15.6.1. ICMode

Container	ICMode	
Description	This container contains the configuration (parameters) of the QDT IC mode.	
Screenshot	Name ICMode	
Properties	**Property**	Value
	Type	IDENTIFIABLE

3.2.1.1.15.6.1.1. EdgeCaputure

Container	EdgeCaputure	
Description	EN: Select which edge to capture.	
Screenshot	channel edge capture QDT_CAPTURE_RISING_EDGE	
Properties	**Property**	Value
	Type	ENUMERATION
	Symbolic Name	none
	Default	QDT_CAPTURE_RISING_EDGE
	Origin	Flagchip

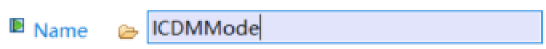
3.2.1.1.15.6.1.2. CVSource

Container	CVSource	
Description	FALSE: capture POSCNT into the CV register TRUE: capture REVCNT into the CV register.	
Screenshot	capture CV from REVCNT	
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.15.6.1.3. ResetPOSCNT

Container	ResetPOSCNT	
Description	FALSE:QDT counter is not reset when the selected channel (n) input event is detected.	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.15.6.2. ICDMMode

Container	ICDMMode	
Description	This container contains the configuration (parameters) of the QDT ICDM mode.	
Screenshot		
Properties	**Property**	Value
	Type	IDENTIFIABLE

3.2.1.1.15.6.2.1. HighActive

Container	HighActive	
Description	select high /low valid of input signal.	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.15.6.2.2. continueMeasure

Container	continueMeasure	
Description	Select whether continue to measure.	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false

	Origin	Flagchip

3.2.1.1.15.6.2.3. StartWithActive

Container	StartWithActive	
Description	TRUE: the channel starts measuring after the first edge is detected. FALSE: the measurement starts immediately after activating the channel	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.15.6.3. ICPMMode

Container	ICPMMode	
Description	This container contains the configuration (parameters) of the QDT ICPM mode.	
Screenshot		
Properties	**Property**	Value
	Type	IDENTIFIABLE

3.2.1.1.15.6.3.1. EdgeCaputure

Container	EdgeCaputure	
Description	EN: Select which edge to capture.	
Screenshot		
Properties	**Property**	Value
	Type	ENUMERATION
	Symbolic Name	none
	Default	QDT_CAPTURE_RISING_EDGE
	Origin	Flagchip

3.2.1.1.15.6.3.2. continueMeasure

Container	continueMeasure	
Description	Select whether continue to measure.	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN

	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.15.6.3.3. StartWithActive

Container	StartWithActive	
Description	TRUE: the channel starts measuring after the first edge is detected. FALSE: the measurement starts immediately after activating the channel	
Screenshot	start with active	
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.15.6.4. ICENMMode

Container	ICENMMode	
Description	This container contains the configuration (parameters) of the QDT ICENM mode.	
Screenshot	Name ICENMMode	
Properties	**Property**	Value
	Type	IDENTIFIABLE

3.2.1.1.15.6.4.1. EdgeCaputure

Container	EdgeCaputure	
Description	EN: Select which edge to capture.	
Screenshot	channel edge capture QDT_CAPTURE_RISING_EDGE	
Properties	**Property**	Value
	Type	ENUMERATION
	Symbolic Name	none
	Default	QDT_CAPTURE_RISING_EDGE
	Origin	Flagchip

3.2.1.1.15.6.4.2. StartPoint

Container	StartPoint
Description	start-point of measure window.
Screenshot	Start Point of window (0 -> 4294967295)

Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	0
	Origin	Flagchip

3.2.1.1.15.6.4.3. EndPoint

Container	EndPoint	
Description	end-point of measure window.	
Screenshot		
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	0
	Origin	Flagchip

3.2.1.1.15.6.4.4. continueMeasure

Container	continueMeasure	
Description	Select whether continue to measure.	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.15.6.5. ICEXPENMMode

Container	ICEXPENMMode	
Description	This container contains the configuration (parameters) of the QDT ICEXPENM mode.	
Screenshot		
Properties	**Property**	Value
	Type	IDENTIFIABLE

3.2.1.1.15.6.5.1. EdgeCaputure

Container	EdgeCaputure
Description	EN: Select which edge to capture.

Screenshot		
Properties	**Property**	Value
	Type	ENUMERATION
	Symbolic Name	none
	Default	QDT_CAPTURE_RISING_EDGE
	Origin	Flagchip

3.2.1.1.15.6.5.2. ExpectedNum

Container	ExpectedNum	
Description	a expected number (ICEXP_NUM) of input edges.	
Screenshot		
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	0
	Origin	Flagchip

3.2.1.1.15.6.5.3. continueMeasure

Container	continueMeasure	
Description	Select whether continue to measure.	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.15.6.6. QUADMode

Container	QUADMode	
Description	This container contains the configuration (parameters) of the QDT QUAD mode.	
Screenshot		
Properties	**Property**	Value
	Type	IDENTIFIABLE

3.2.1.1.15.6.6.1. NormalPolarity

Container	NormalPolarity

Description	Select PHA or PHB normal porlarity in QUAD mode. true: normal porlarity. false: inverted porlarity.	
Screenshot	use normal polarity	
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.1.1.15.6.6.2. CVValue

Container	CVValue	
Description	CV is used to match event. Compare with the source selected by MatchType	
Screenshot	CV value for match event (0 -> 4294967295) 0	
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	0
	Origin	Flagchip

3.2.1.1.15.6.6.3. MatchType

Container	MatchType	
Description	EN: Select the match type for CV.	
Screenshot	match type QDT_CHANNEL_MATCH_DISABLE	
Properties	**Property**	Value
	Type	ENUMERATION
	Symbolic Name	none
	Default	QDT_CHANNEL_MATCH_DISABLE
	Origin	Flagchip

3.2.2. QDTGeneral

Container	QDTGeneral	
Description	This container holds the parameters related each QDT Driver Unit.	
Screenshot	QDTGeneral Name QDTGeneral	
Properties	**Property**	Value
	Type	IDENTIFIABLE

3.2.2.1. QDTDevErrorDetect

Container	QDTDevErrorDetect	
Description	Switches the Development Error Detection and Notification: ON or OFF. When this option is OFF code size is reduced, but no error detection is available.	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.2.2. QDTMulticoreSupport

Container	QDTMulticoreSupport	
Description	Switches multicore support on or off: False: For all variants, no EcucPartition shall be referenced in QDTEcucPartitionRef. True: For all variants, at least one EcucPartition needs to be referenced in QDTEcucPartitionRef.	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.2.3. QDTVersionInfoApi

Container	QDTVersionInfoApi	
Description	Switches the QDT_GetVersionInfo() API: ON or OFF. When this option is ON driver supports API for getting Version information for the Driver.	
Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	false
	Origin	Flagchip

3.2.2.4. QDTDeInitApi

Container	QDTDeInitApi
Description	Switches the QDT_DeInit() API: ON or OFF. When this option is ON driver supports API for getting Version information for the Driver.

Screenshot		
Properties	**Property**	Value
	Type	BOOLEAN
	Symbolic Name	false
	Default	true
	Origin	Flagchip

3.2.2.5. QDTIndex

Container	QDTIndex	
Description	Specifies the InstanceId of this module instance. If only one instance is preQDT it shall have the Id 0.	
Screenshot		
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	0
	Origin	Flagchip

3.2.2.6. QDTEcucPartitionRef

Container	QDTEcucPartitionRef	
Description	Maps the QDT driver to zero or multiple ECUC partitions to make the module's API available in this partition. The QDT driver will operate as an independent instance in each of the partitions.	
Screenshot		
Properties	**Property**	Value
	Type	REFERENCE
	Symbolic Name	none
	Default	none
	Origin	Flagchip

3.2.3. CommonPublishedInformation

Container	CommonPublishedInformation	
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.	
Screenshot		
Properties	**Property**	Value
	Type	IDENTIFIABLE

3.2.3.1. ArReleaseMajorVersion

Container	ArReleaseMajorVersion	
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.	
Screenshot		
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	4
	Origin	Flagchip

3.2.3.2. ArReleaseMinorVersion

Container	ArReleaseMinorVersion	
Description	Minor version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	6
	Origin	Flagchip

3.2.3.3. ArReleaseRevisionVersion

Container	ArReleaseRevisionVersion	
Description	Revision version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	0
	Origin	Flagchip

3.2.3.4. ModuleId

Container	ModuleId
Description	Module ID of this module from Module List. Note: Implementation Specific Parameter
Screenshot	

Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	255
	Origin	Flagchip

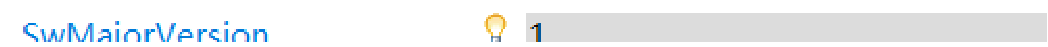
3.2.3.5. SwMajorVersion

Container	SwMajorVersion	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific. Note: Implementation Specific Parameter	
Screenshot		
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	1
	Origin	Flagchip

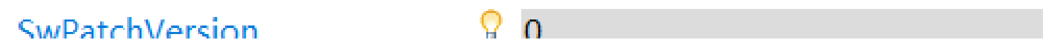
3.2.3.6. SwMinorVersion

Container	SwMinorVersion	
Description	Minor version number of the vendor specific implementation of the module. The numbering is vendor specific. Note: Implementation Specific Parameter	
Screenshot		
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	1
	Origin	Flagchip

3.2.3.7. SwMajorVersion

Container	SwMajorVersion	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific. Note: Implementation Specific Parameter	
Screenshot		
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	1
	Origin	Flagchip

3.2.3.8. SwPatchVersion

Container	SwPatchVersion	
Description	Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific. Note: Implementation Specific Parameter	
Screenshot		
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	1
	Origin	Flagchip

3.2.3.9. VendorApiInfix

Container	VendorApiInfix	
Description	In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.	
Screenshot	n/a	
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	none
	Origin	Flagchip

3.2.3.10. VendorId

Container	VendorId	
Description	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list. Note: Implementation Specific Parameter	
Screenshot		
Properties	**Property**	Value
	Type	INTEGER
	Symbolic Name	false
	Default	none
	Origin	Flagchip

Chapter 4. Configuration Guides

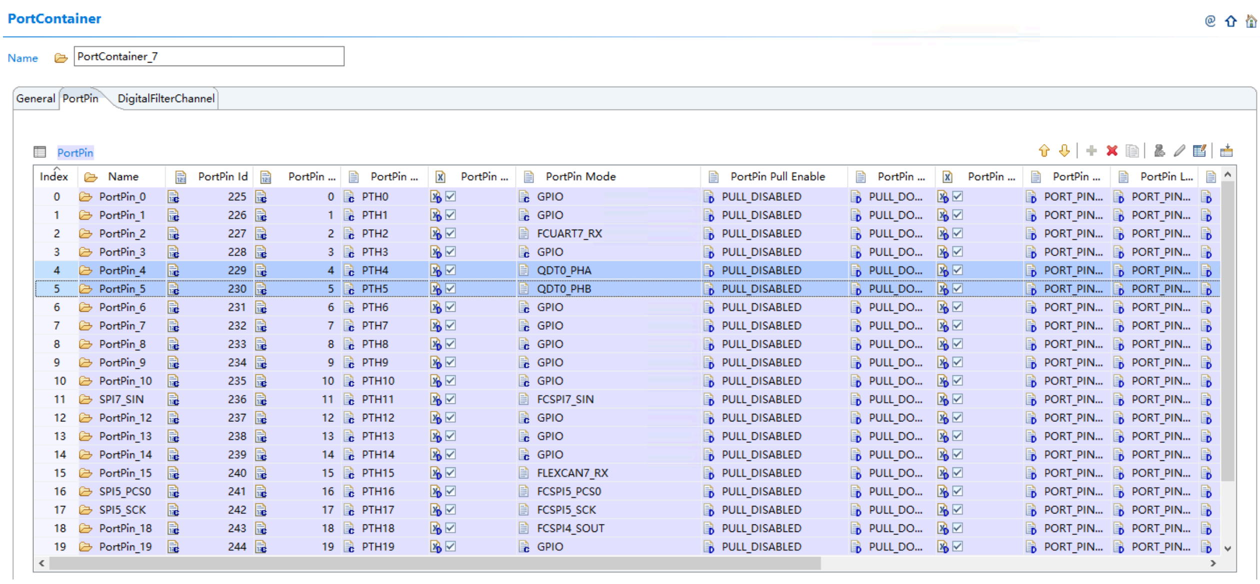
4.1. QDT Usage Common Steps

Basically, the Qdt module can be configured in 4 steps:

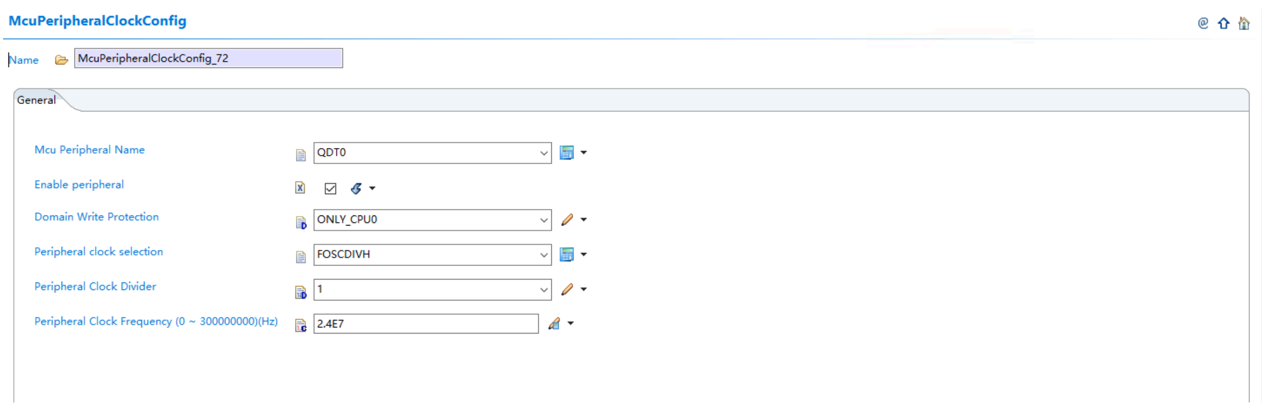
- Configure Qdt general configurations and enable Qdt Port&clock configuration.
- Configure Qdt controller configuration.
- Configure Qdt channel configuration
- Generate configuration files.

4.2. QDT Usage Demo

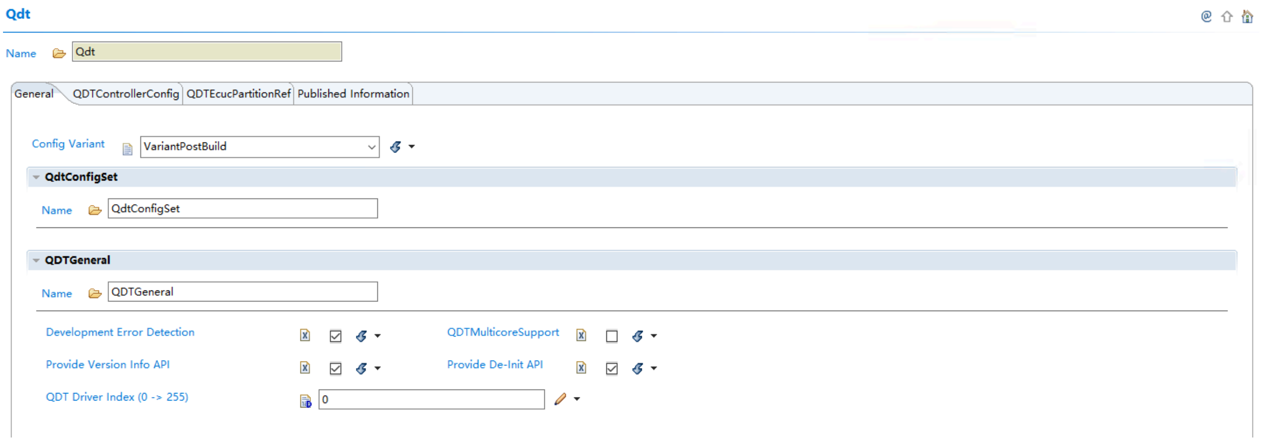
1. Users should set the Qdt pin mode in PortContainer firstly.



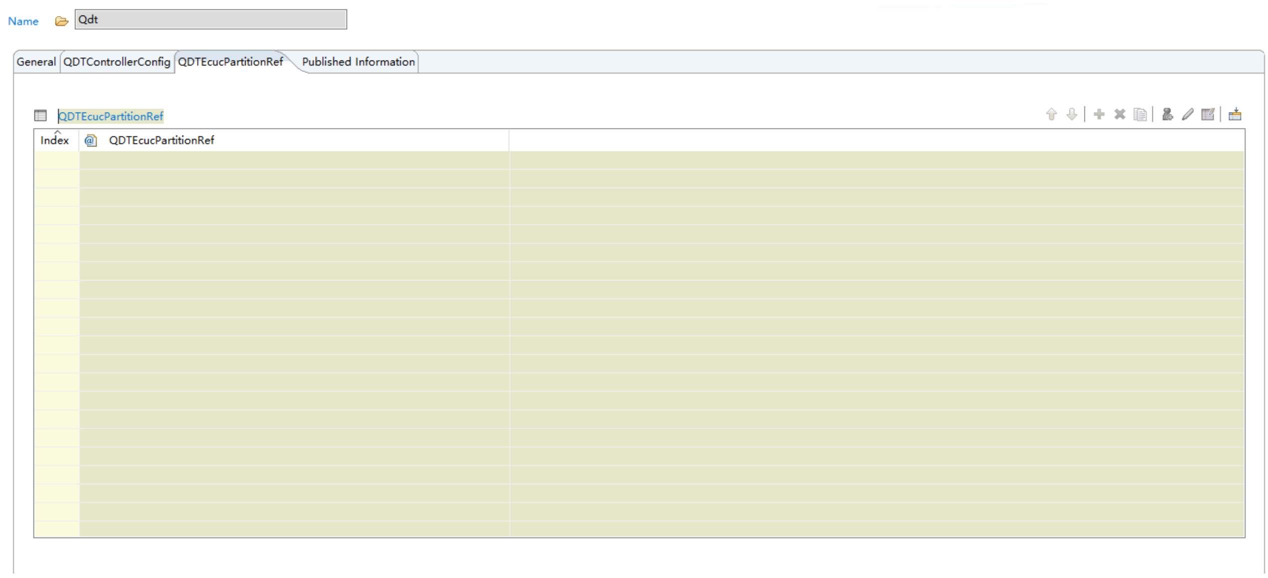
2. Then set the Qdt clock in McuClockReferencepoint_PCC container.



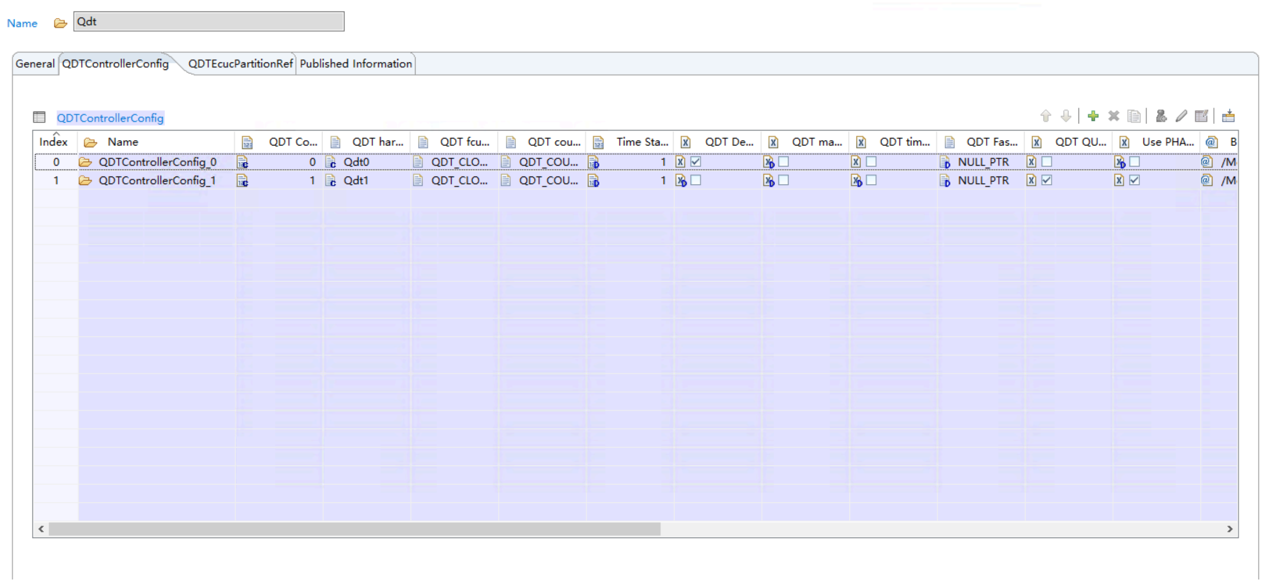
3. In Qdt General configuration tag, users can configure if QdtMulticoreSupport is used. If Det isn't needed, the Development Error Detection checkbox has no need to be checked.

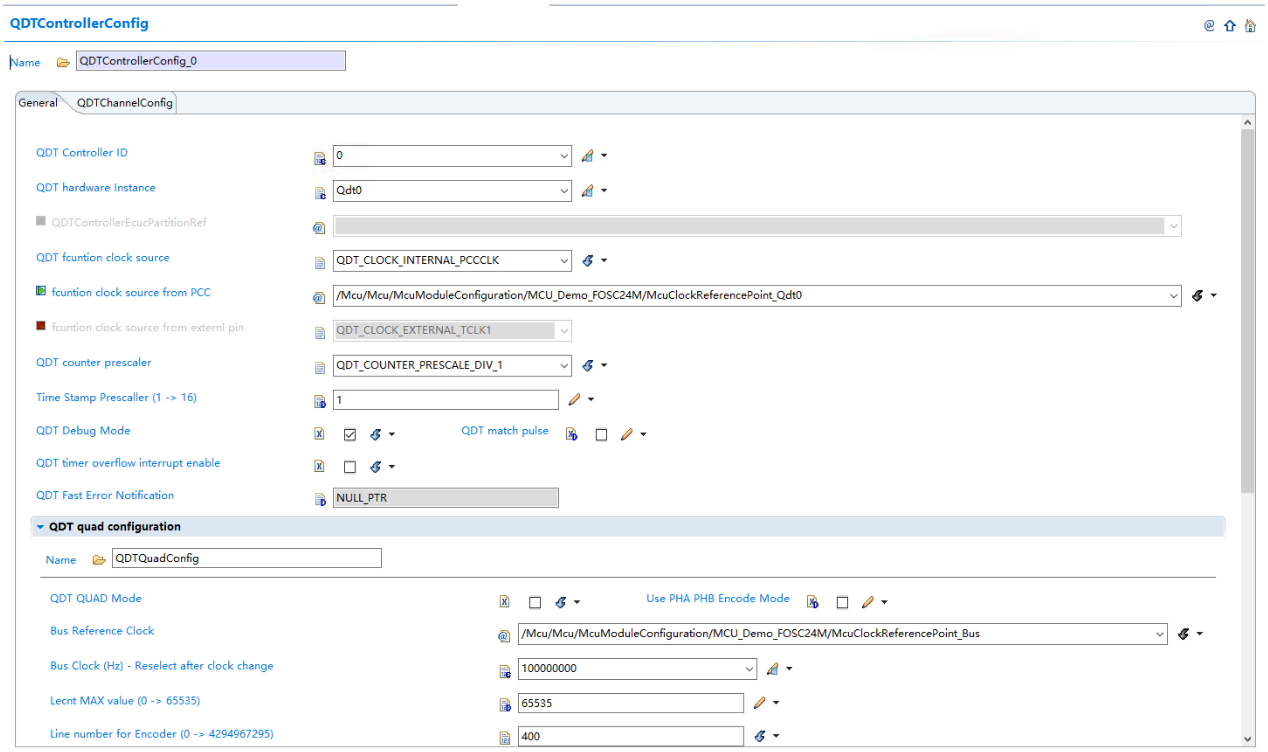


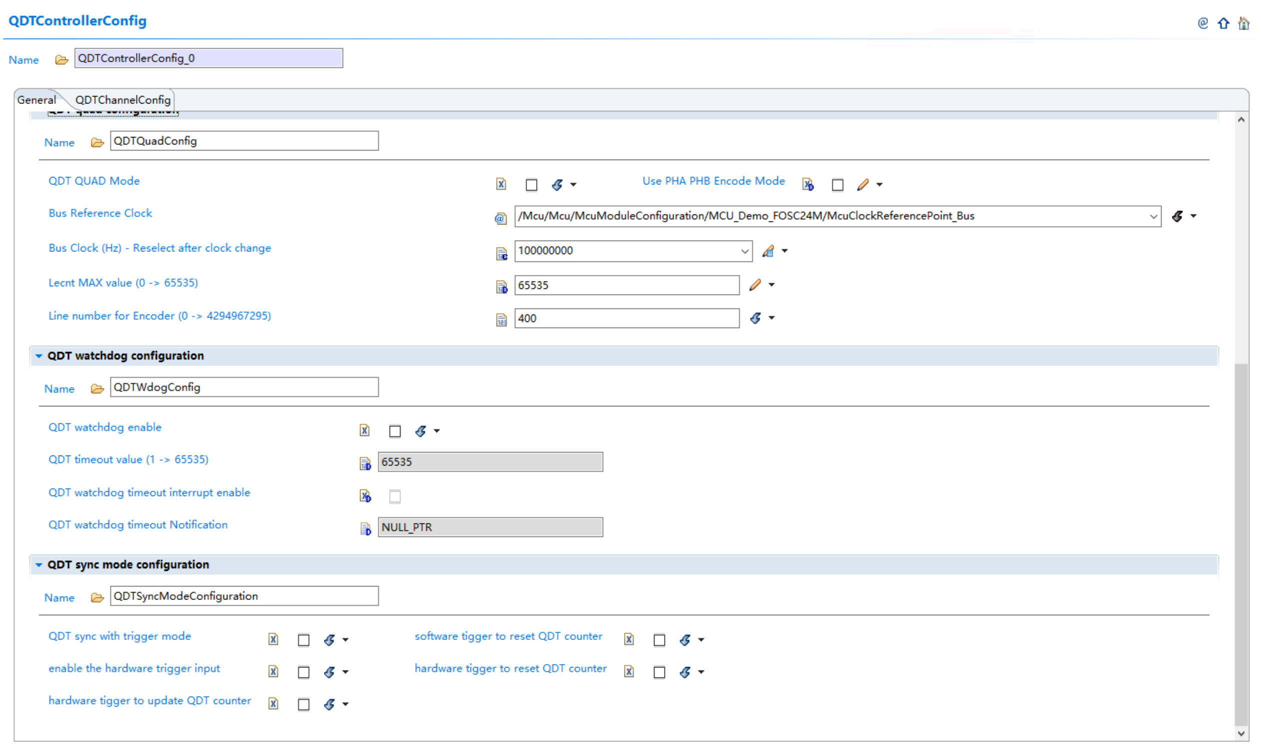
4. In QdtEcucPartitionRef tag, users can select what cores would use the Qdt controller.



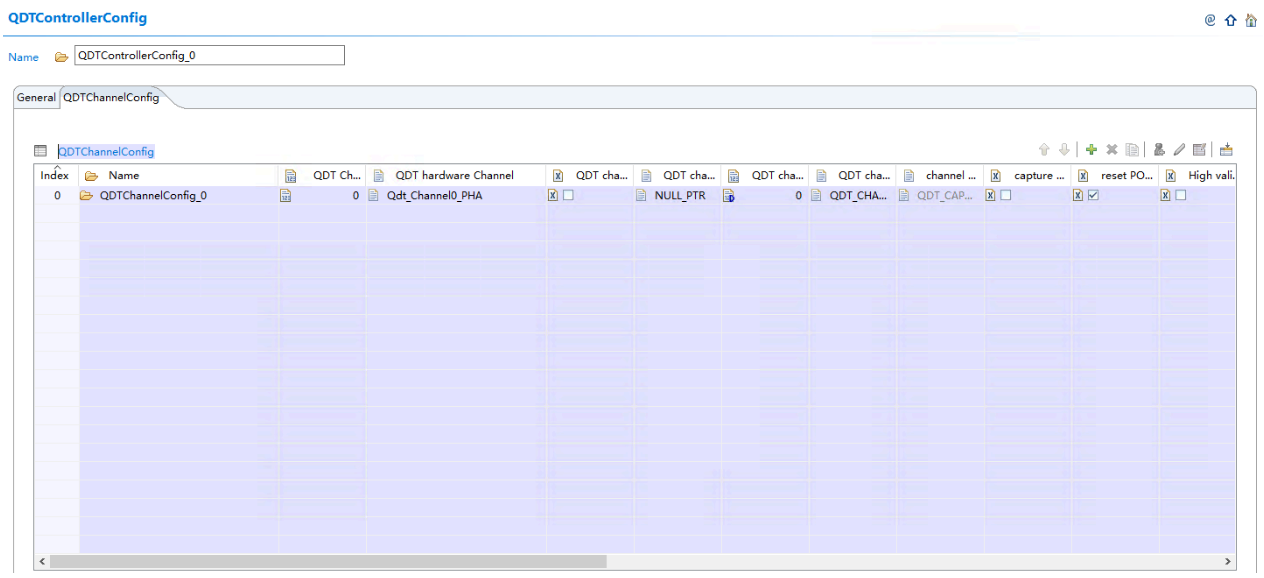
5. In QdtControllerConfig tag, users can select what Qdt controller would be configured and used.

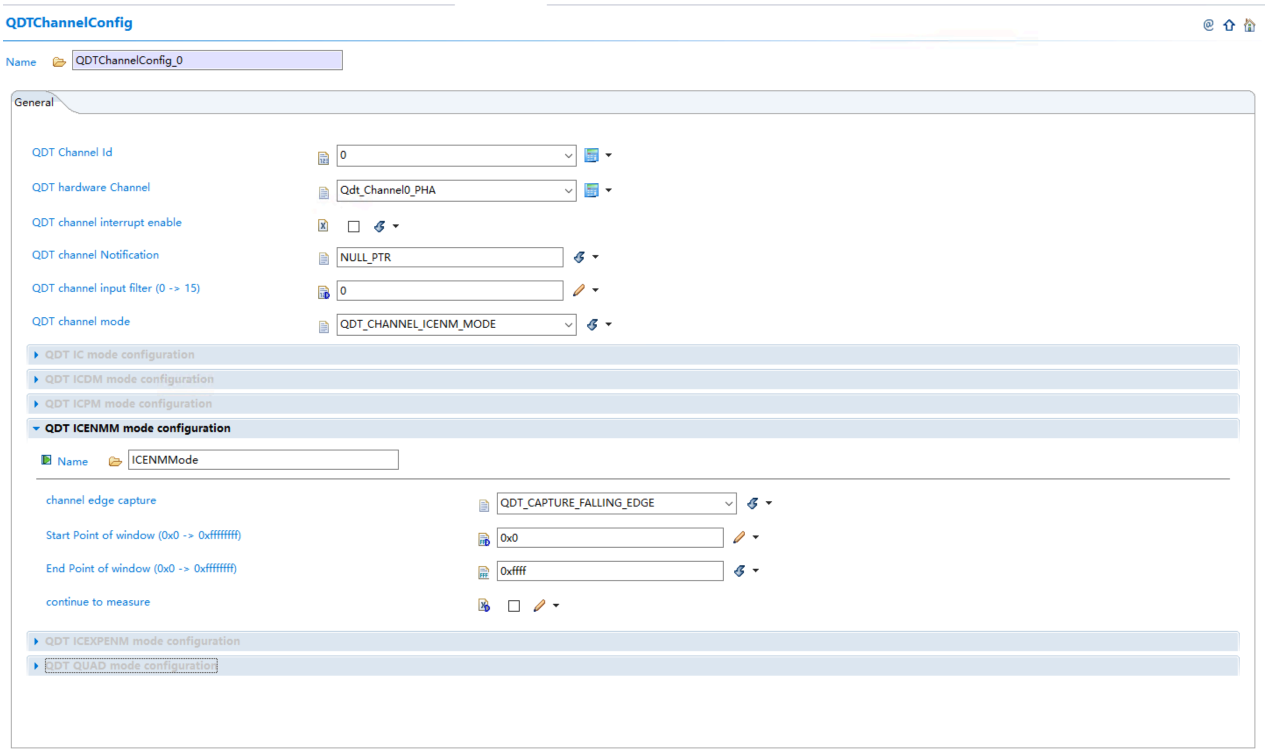




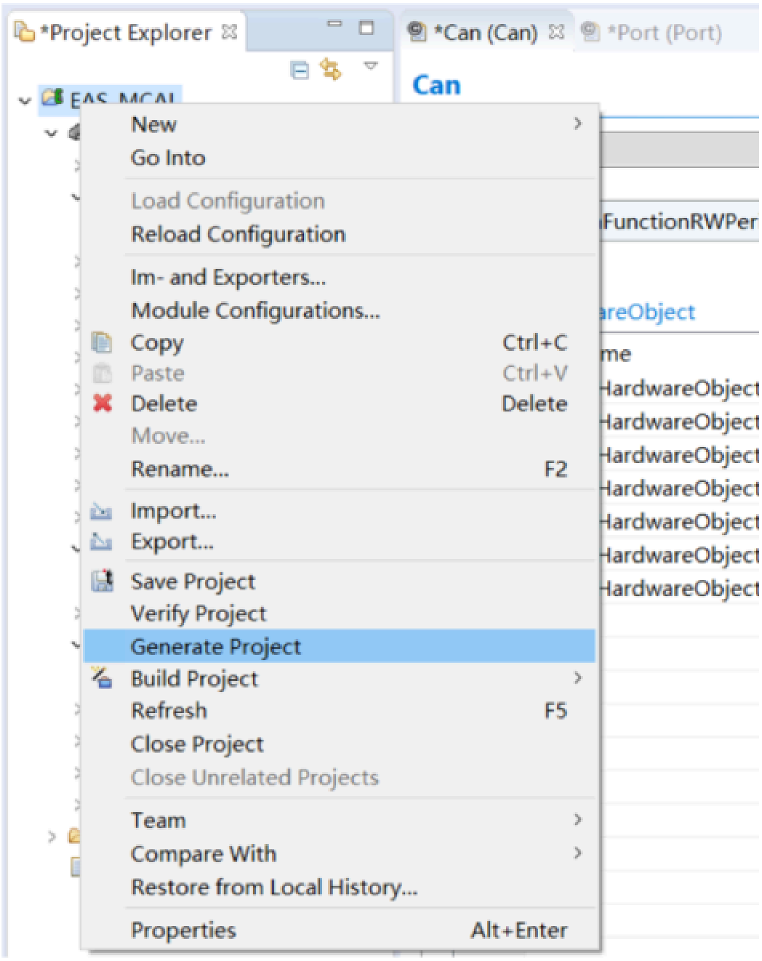


6. In QdtChannelConfig tag, users can select what Qdt channel would be configured and used.





7. Click the Generate Project to generate relevant code.


*Project Explorer
*Can (Can)
*Port (Port)
Can
New
Go Into
Load Configuration
Reload Configuration
Im- and Exporters...
Module Configurations...
Copy Ctrl+C
Paste Ctrl+V
Delete Delete
Move...
Rename... F2
Import...
Export...
Save Project
Verify Project
Generate Project
Build Project
Refresh F5
Close Project
Close Unrelated Projects
Team
Compare With
Restore from Local History...
Properties Alt+Enter